

Titration simulation of varying half-life medications

Objective

The half-life of a drug corresponds to the time it takes for the concentration of the medication in the blood plasma to reduce by half after administration has been stopped. It is directly dependent on the volume of distribution and inversely dependent on the blood clearance level of the patient.

$$t_{1/2} = \frac{\ln(2) \cdot \text{Volume of distribution}}{\text{Clearance}} \quad (1)$$

This equation establishes that the ratio between the volume of distribution and clearance level determines half-life. It then follows that for drugs that feature a shorter half-life, the larger ratio of clearance level to volume of distribution implies that this drug will reach a steady state concentration faster than a drug with a comparatively longer half-life due to the combined effects of efficient clearing and a subsequently faster response to change in dosing regimen. Conversely, drugs with longer half-lives, and thus larger ratios of volume of distribution to clearance level, will feature greater albeit slower accumulation of the drug in the body as a greater proportion of the drug remains in the body after each time step. Therefore, drugs with shorter half-lives are expected to reach a steady-state concentration faster than drugs with longer half-lives, and this steady-state concentration will be on average lower than the steady-state concentration of a drug with a longer half-life.

We aim to demonstrate this effect using pharmacokinetic equations and modelling.

One-compartment Model

Relevant Equations

The most basic method of modelling the dynamics of drug input and elimination is the one compartment model. The critical assumption is that the entire body comprises one single central compartment with immediate and uniform distribution of the drug. In our model, we will consider this compartment to be the blood plasma. We will assume for simplicity that only one drug is being administered. We will neglect factors of biological variation such as metabolism which may affect rates of drug accumulation and elimination in individuals.

With these conditions, at any time t , the concentration of the drug within the central compartment will be determined by two factors: the dose rate and the rate of decay due to elimination from the body. This decay is dependent on the half-life of the drug. Then, we arrive at the traditional pharmacokinetic equation for drug concentration over time with a continuous infusion, where the constant k represents the rate of elimination of the drug from the central compartment.

$$\text{Plasma Concentration at time } n = C_p(n) = \frac{\text{Infusion rate}}{\text{Clearance}} \cdot (1 - e^{-kt}) \quad (2)$$

We use this continuous equation to develop a discrete-time recursive model in which we can sample drug concentrations at specified intervals of time, as might happen in a clinical or experimental scenario. This requires us to approximate the equation above by iteratively updating the drug concentration at each time step, and accounting for both the input of the drug from the ongoing infusion as well as the decay factor, which will eliminate a portion of the residual drug concentration from the previous time step.

The resulting equation can be further refined to reflect these processes.

$$C_p(n) = \text{Drug input at time step } n + \text{Remaining drug from } C_p(n-1) \quad (3)$$

To account for the drug input, we will want to consider the amount of drug that is delivered per time step. We achieve this by taking the product of the concentration of the drug within the solution being infused and the infusion flow rate. Then, we adjust this value to amount of drug being delivered within the sampling interval by multiplying by the time t at step n . Then, to account for difference in blood volume between individuals which would affect the concentration of the drug between individuals, we will divide by the volume of distribution of the central compartment. To account for the decay of the residual drug, we want to use a decay factor which takes into account the rate of elimination, which can be expressed as $\frac{\ln(2)}{\text{Drug half-life (min)}}$. Adjusting our equation to convert half-life to minutes in order to align its units with that of the sampling interval, we get the following equation.

$$G = e^{\frac{-T_{\text{samp}}}{t_{\text{half}} \frac{\ln(2)}{\ln(2)}}} \quad (4)$$

T_{samp} is the unit of time seconds, t_{half} is the drug single-compartment half-life in seconds.

$$C_p[n \cdot T_{\text{samp}}] = \frac{T_{\text{samp}} \cdot \text{FlowRate}[n \cdot T_{\text{samp}}] \cdot \text{Conc}}{V_{\text{dist}}} + G \cdot C_p[(n-1) \cdot T_{\text{samp}}] \quad (5)$$

$C_p[n \cdot T_{\text{samp}}]$ is the instantaneous concentration in units of drug per liter of plasma; V_{dist} is the calculated effective distribution volume of drug based on dose and measured concentration in a population; Conc is the number of units of the drug per ml at which it is administered;

Flow Rate is the number of ml administered per unit time.

$$C_{p,SS} = \frac{T_{\text{samp}} \cdot \text{MaxFlowRate} \cdot \text{Conc}}{1 - G} \quad (6)$$

MaximumFlowRate is the chosen peak flow of 32 ml/h for this study; Conc is the number of units of the drug per ml at which it is administered; G is the half-life and sample period dependent gain constant.

$$C_{p, \text{Normalized}}(t) = \frac{C_p(t)}{C_{p, \text{Steady-State}}(t)} \times 100 \quad (7)$$

For drugs with longer half-lives, the decay factor G in equations (4) and (5) is closer to 1, meaning the calculated drug concentration will be less responsive to changes in dosing. In this real-time model, this indicates to us that the half-life is key to predicting how fast concentration will change in response to a clinician's actions. For shorter half-life drugs, dosing adjustments will quickly alter concentration levels, necessitating more frequent dosing whilst remaining vulnerable to potential overshooting or undershooting. For longer half-life drugs, dosing adjustments would lead to slow changes in the plasma concentration; however, clinicians would need to remain diligent of the fact that frequent changes in dosing adjustments administered in interest of overcoming delayed effects in the short-term would potentially lead to sustained high concentration due to the accumulation in the long-term.

R, the ratio of step-duration to half-life

We introduce the parameter R as a relevant metric for titration protocol design. We suggest that keeping $R > 0.3$ allows for near 100% achievement of target normalized concentration for a given step duration.

Referenced Drugs

To examine the effect of varying half-lives and referenced clearances, we will utilize and titrate a range of drugs with known pharmacokinetic properties. These are chiefly hemodynamic and vasoactive agents common in clinical practice and particularly in dynamic environments in which a clinician may be clamping or unclamping major vessels, weaning a patient from cardiopulmonary bypass, or actively responding to acute pump failure or hemorrhage during an operation. The titration of infusion rates corresponds to the flow rates used to infuse the drug in clinical settings, the concentration of the drug used at these flowrates was similarly referenced. Clearance and volume of distribution for each drug was chosen within the range of referenced values and kept constant between individuals. These theoretical drug parameters were chosen to produce variation in duration of half-lives.

1. **Clevidipine** Clevidipine is a calcium channel blocker and vasodilator used to treat hypertension in patients, commonly seen as Cleviprex. Clevidipine has a half-life of one minute, and as a result, it is commonly used to rapidly treat acute cases of elevated blood pressure [5]. The clearance ranges from 0.069 to 0.14 L/min/kg [11] and the volume of distribution ranges from 0.19 to 0.6 L/kg. [10] The concentration of clevidipine when infused is 0.5 mg per ml of diluent [5], and the molar mass of clevidipine is 456.03 g/mol. [4]

2. **Nitroglycerin** Nitroglycerin is a vasodilatory and venodilatory drug used to treat hypertension as well as inducing hypotension. [14] Nitroglycerin when administered intravenously has a half-life of approximately 2 minutes. [14] The clearance ranges from 0.3 to 1 L/min/kg [15] and the volume of distribution is ranges from 2.1 to 4.5 L/kg. [17] The concentration of nitroglycerin when infused at the chosen flow rates is about 400 mcg per ml of diluent, and the molar mass of nitroglycerin is 227.09 g/mol. [17]

3. **Norepinephrine** Norepinephrine, or norepinephrine, is a sympathomimetic drug commonly used to rapidly treat patients with severe hypotension. The desired effects are chiefly achieved by an increase in vascular and venous resistance [12], which in turn increases, and helps stabilize, blood pressure. Norepinephrine is commonly marketed as Levophed. [20] The drug has a half-life between 2.5 to 5 minutes [18] a clearance of approximately 1.4 L/min/kg [13] and a volume of distribution of about 8 L/kg [21]. The concentration of norepinephrine when infused at the chosen flow rates is about 32 mcg per ml of diluent, and the molar mass of the compound is about 0.169 ug/mol [19].

3. **Epinephrine** Epinephrine is an antihypotensive and vasoconstrictor drug used to treat low blood pressure as well as anaphylactic shock in patients. Due to these characteristics, epinephrine has an important role in reversing cardiac arrest and immediate administration is associated with better recovery in patients [8]. Epinephrine has a half-life between 3-10 minutes [24], with a clearance of approximately 0.147 L/kg/min [22]. The clearance ranges between 0.048 to 0.056 L/min/kg [6], and the volume of distribution ranges between 0.5 to 3.5 L/kg [3]. The molar mass of epinephrine is approximately 0.183 g/mol, [9] and the concentration of epinephrine when administered at these flow rates is 32 mcg per ml of diluent.

4. **Vasopressin** Vasopressin, or ADH/AVP, is a vasopressor agent used to elevate blood pressure levels in patients that are in a state of vasodilatory shock, which may occur due to sepsis, anaphylaxis, or following cardiac surgery [1] [16]. Vasopressin is also extremely useful when administered to treat cardiac arrest, which can result in vasodilatory shock [7]. Vasopressin has a half-life ranging between 10 to 35 minutes, a clearance of approximately 0.01 to 0.02 L/kg/min, and a volume of distribution of 0.14 L/kg [16] The concentration of vasopressin when administered at the chosen flow rates is typically 1U per ml of diluent,

which is equivalent to 1.88 mg per ml of diluent [23] The molar mass of vasopressin is 1.028 ug/nmol [2].

Results

When modeling a one-compartment model, we would expect to see the demonstration of two key principles in pharmacokinetic dynamics. Between all patients, shorter half-lives should correspond to a faster achievement of steady-state concentrations due to rapid adjustment to dosing rate changes. The ratio of drug clearance and volume of distribution will chiefly determine how quickly the drug achieves steady state. Combining these two effects, we can consider two hypothetical drugs which share the same ratio of clearance to volume of distribution (implying equivalence in half-life duration). While both drugs will feature the same degree of responsiveness to changes in dosing regimen, the drug with the larger volume of distribution will have a higher steady-state plasma concentration when administered at the same dosing rate as the other drug. This effect should be apparent when we compare specific drug concentration over time between patients of varying weights and subsequently varying volumes of distribution. We use equation (5) to model theoretical drug concentration over time for three patients of varying weights, where dosing adjustments are made according to the varying time-step intervals in the figures below.

We see the expected results in the final figures¹: clevidipine, which features the shortest half-life and small clearance, rapidly achieves steady-state level and clears efficiently as dose decreases. Nitroglycerin closely follows the dynamics of Clevidipine due to its similarity in half-life. It is less responsive to change in dosing rate than Clevidipine due to its smaller ratio of clearance to volume of distribution, and also features a lower steady-state concentration due to its comparatively larger volume of distribution. This effect is repeated for norepinephrine and vasopressin when compared respectively to Clevidipine and nitroglycerin.

With regard to volume of distribution, Clevidipine has the smallest volume of distribution and a moderate clearance. As such, clevidipine features greater accumulation of the drug within the body at each time step when compared to the other drugs. Nitroglycerin and norepinephrine have larger volumes of distributions as compared to their clearances, which results in much slower accumulation and lower troughs in concentration as well as less efficient clearance for both drugs. Epinephrine has a smaller volume of distribution than both, which allows it to increase in concentration more rapidly within the central compartment as compared to norepinephrine, despite featuring slower responsiveness to dosing changes. This is due to the fact that epinephrine has a significantly lower volume of distribution, but also a clearance that is smaller in magnitude. This contributes to its larger half-life. Vasopressin, despite having an equivalent volume of distribution to Clevidipine, also features a much lower clearance level. These two effects combined mean that the concentration of Vasopressin is significantly less responsive to changes in dose ($G \gg 1$), which makes drug accumulation slow. This contrasts with Clevidipine, which was able to accumulate much faster due to its efficiency in clearance, allowing it to benefit from a small volume of distribution through increased responsiveness to dose rate. The effect of volume of distribution on drug accumulation is easier to see when comparing between patients within a specific drug: patients with larger weights have larger volumes of distributions, and thus adult patients will have less accumulation of the drug over time when compared to adolescents and infants, the latter of whom will tend to feature rapid accumulation of the drug due to their significantly smaller blood volume.

To summarize, drugs with shorter half-lives, that is, smaller ratios of clearance to volume of distribution, will reach steady state concentration faster due to increased responsiveness to dosing changes and smaller magnitude of G , our decay factor defined by equation (4). Drugs with larger volumes of distribution will feature slower accumulation of the drug in the compartment over time. Drugs with a larger clearance will feature more rapid elimination of the drug over time. Considering these effects in tandem can allow us to predict how drugs will behave in general over time and in response to changes to dosing regimen, not accounting for biological variation between individuals.

Two-compartment model

The one-compartment model is a simple model which is best suited for drugs that behave in a manner similar to the key assumptions the model relies on: approximately uniform and instantaneous distribution with limited diffusion into other tissues. For other drugs, it becomes more pertinent for us to progress to another basic model which accounts for distribution of the drug from the central compartment and into a peripheral compartment. This is the two-compartment model, with which we can estimate concentration in both compartments at each time step using a numerical solution.

¹To be attached...

First, we are interested in modeling how the concentration of the drug in both compartments will change over time using coupled differential equations. We begin with the assumption that the drug is being injected/infused directly into the central compartment, which we will again consider to be the blood plasma.

Then, we can estimate the change in concentration over time in the blood plasma as the total of the concentration of the drug being infused in addition to the amount of drug being redistributed into the central compartment from the peripheral compartment at time t . From this we will need to subtract the portion of the drug in the central compartment that will be eliminated out of the blood plasma and body due to the elimination constant k at time t , as well as the portion of the drug in the central compartment that is being distributed into the peripheral compartment at time t .

$$\frac{d}{dt}[\text{Concentration in central comp.}] = \frac{\text{Dose input}}{V_1} - (k_e + k_{12}) \cdot C_{\text{central}} + k_{21} \cdot C_{\text{peripheral}} \quad (8)$$

The drug is only eliminated from the central compartment. This means that the differential equation modeling change in concentration within the peripheral compartment is simpler and only consists of the input of the drug being distributed in from the central compartment, with the portion of the drug within the compartment being redistributed into the central compartment subtracted from this total.

$$\frac{d}{dt}[\text{Concentration in peripheral compartment}] = k_{12} \cdot C_{\text{central}} - k_{21} \cdot C_{\text{peripheral}} \quad (9)$$

Due to their linearity, we can write these equations as a linear system. This will allow us to use the matrix exponential to solve the system and find concentration of the drug in each compartment at each time t .

$$C(t) = \begin{bmatrix} [\text{Concentration in central compartment}] = C_1(t) \\ [\text{Concentration in peripheral compartment}] = C_2(t) \end{bmatrix} \quad (10)$$

$$\text{System matrix, } A = \begin{bmatrix} -(k_{10} + k_{12}) & k_{12} \\ k_{21} & -k_{21} \end{bmatrix} \quad (11)$$

$$\text{Dose input vector, } B = \begin{bmatrix} \frac{\text{Dose rate}}{V_1} \\ 0 \end{bmatrix} \quad (12)$$

We will express these equations in a first order linear differential system, and we want an approximation solution for concentration $C(t)$ in equation (12) below at each time step $(t + dt)$.

$$\frac{dC}{dt} = A \cdot C + B \quad (13)$$

The homogeneous part of the system has a solution of the form of equation (12) below.

$$C(t) = \exp(A \cdot t) \cdot C(0) \quad (14)$$

Adding the inhomogeneous term, B , which represents dose input to the central compartment, we assume it is constant over short intervals (i.e. the time steps). Then, we apply the basic Euler method to approximate solution as follows below.

$$C(t + dt) = \exp(A \cdot dt) \cdot C(t) + \int_0^{dt} \exp(A \cdot (dt - \tau)) \cdot B d\tau \quad (15)$$

$$\int_0^{dt} \exp(A \cdot (dt - \tau)) \cdot B d\tau = \left(\int_0^{dt} \exp(A \cdot \tau) d\tau \right) \cdot B \quad (16)$$

$$\int_0^{dt} \exp(A \cdot \tau) d\tau \approx dt \cdot I \quad (17)$$

Where I is the identity matrix. We are assuming through this approximation in equation (15) that with small time step dt , the exponential changes slowly enough that we can approximate the integral as dt . Then, using the Taylor series expansion of the matrix exponential, we will use the first-order approximation of the matrix exponential under our assumption of small dt .

$$C(t + dt) = \exp(A \cdot dt) \cdot C(t) + B \cdot dt \quad (18)$$

We use this matrix exponential method to update concentrations for each time step in our simulation numerically using a simple Matlab script. We again are interested in developing a very basic model, derived directly from our earlier one compartment model, meaning our input parameters will essentially be the same with the addition of specific elimination constants which are drug-specific.

We can also simulate how the drugs we referenced in the one compartment model in section one would theoretically behave in a two compartment model when administered to patients of varying weights within a clinically-relevant stationary time-window of one hour. Following our conclusions from a one-compartment model in the results section above, we would again expect that when comparing between patients within drug, there should be an equivalent degree of responsiveness to change in dosing regimen and equivalence in time to reach steady-state. However, patients with smaller volumes of distributions should feature greater accumulation of the drug over this time interval. Comparing between drugs, we would expect that drugs with smaller half-lives feature greater degrees of responsiveness to dose rate as well as less time to reach steady-state concentrations. Drugs associated with smaller volumes of distribution should feature greater overall accumulation of the drug and higher troughs in concentration as a result, unless also paired with a blood clearance level that is relatively small in magnitude, resulting in a larger half-life and less overall accumulation at each time step.

To examine these effects, we will generate a plot comparing compartment-specific concentration of nitroglycerin in patients of varying weights being administered the drug at increasing and then decreasing dose rates over one hour. We will then compare between drugs by comparing this plot featuring identical patients being administered Clevidipine at the same rates and over the same time window.

Results

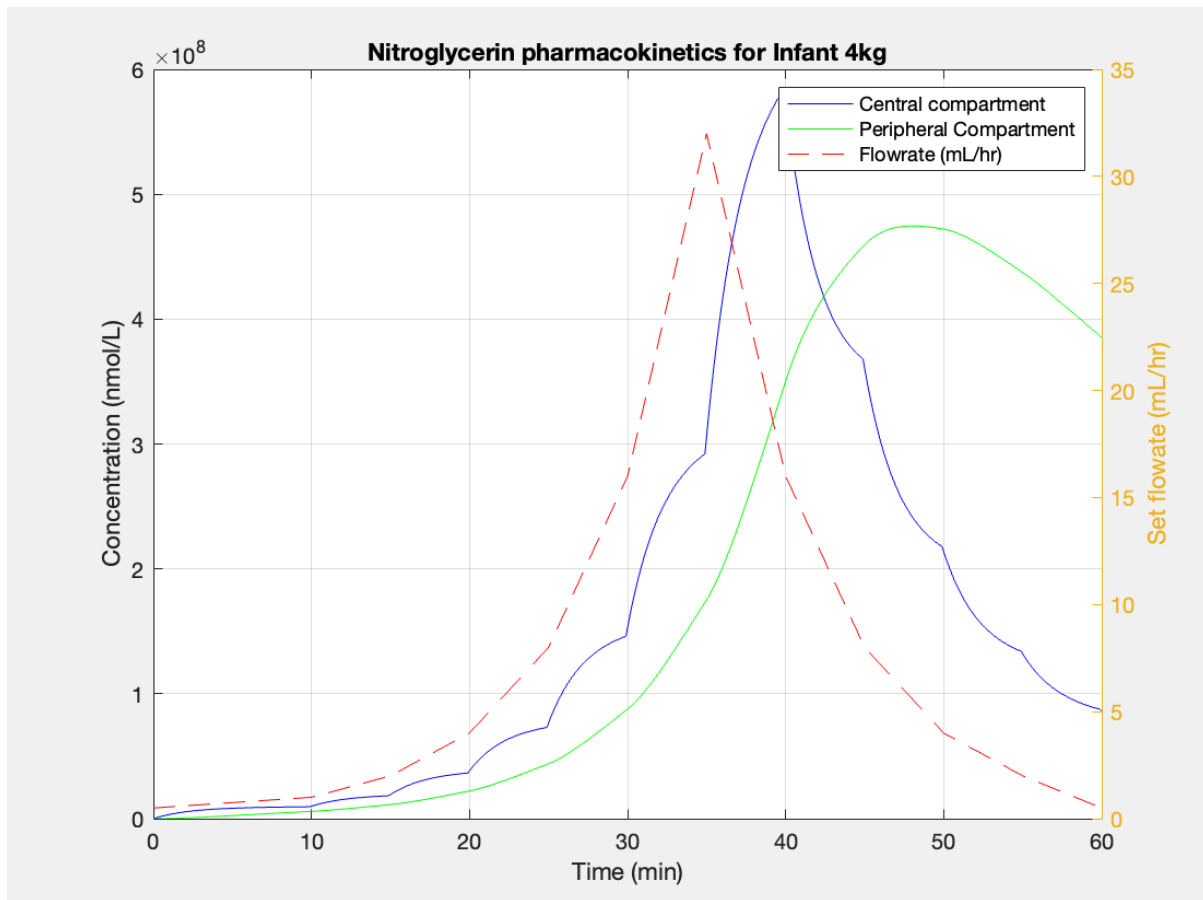


Figure 1: Nitroglycerin concentrations for 4.57 kg patient.

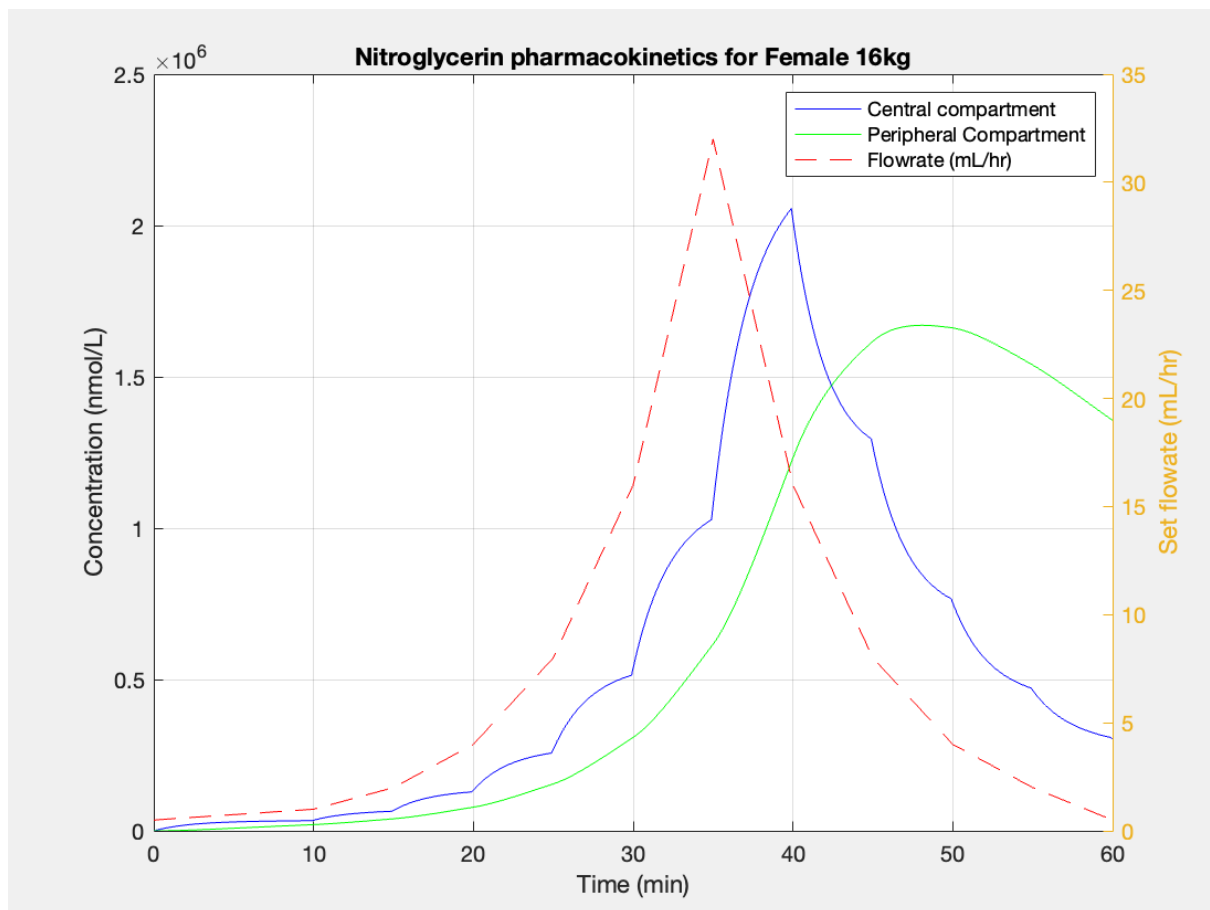


Figure 2: Nitroglycerin concentrations for 18.31 kg patient.

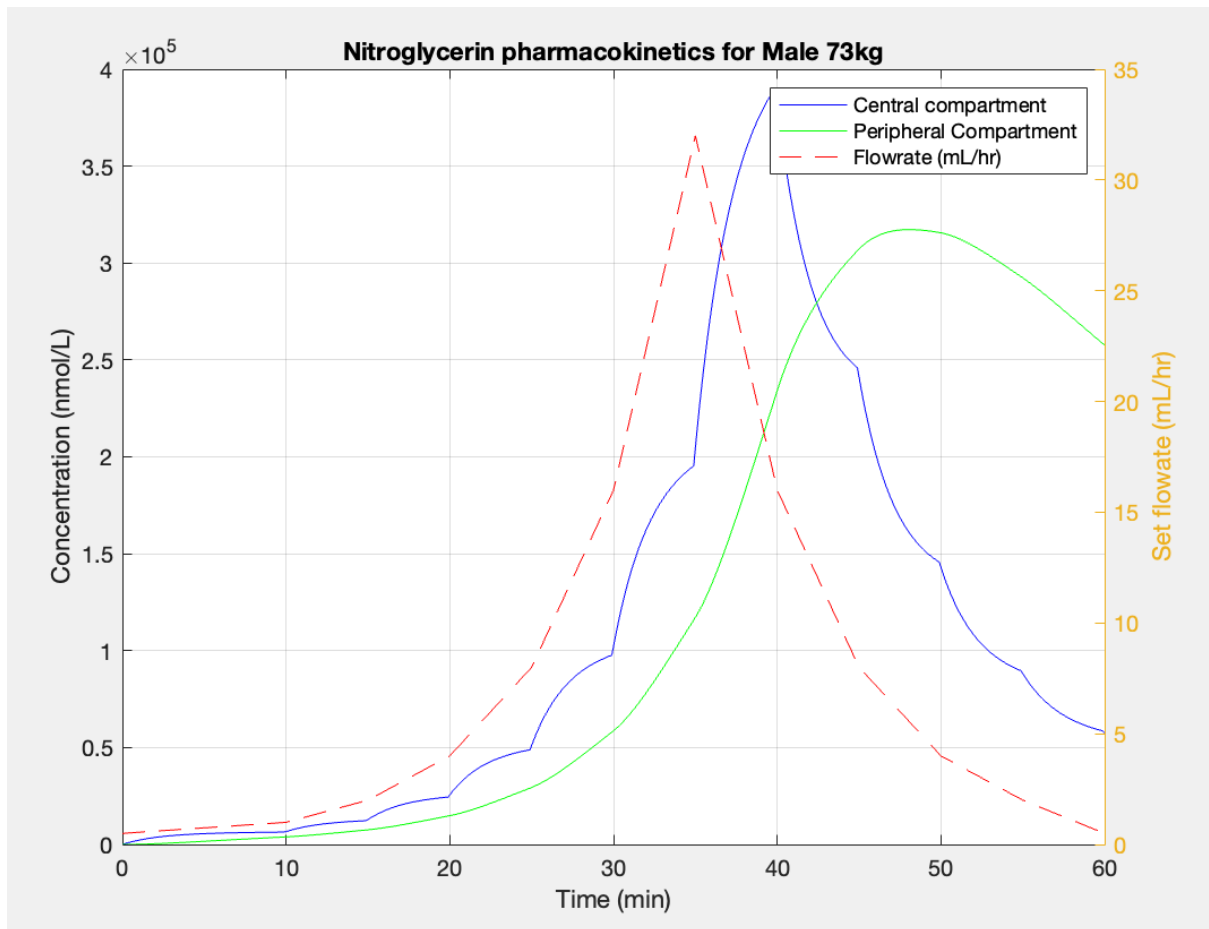


Figure 3: Nitroglycerin concentrations for 73 kg patient.

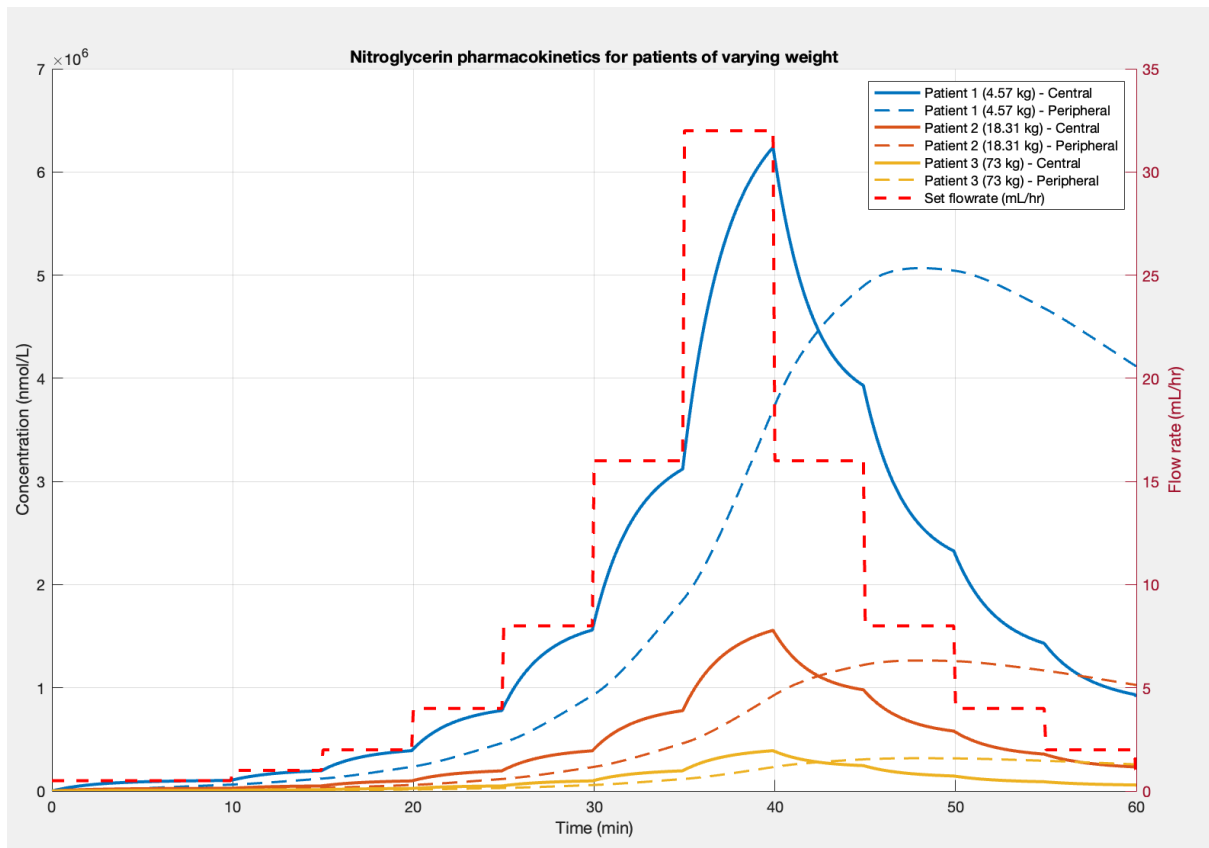


Figure 4: Nitroglycerin pharmacokinetics for patients of varying weight.

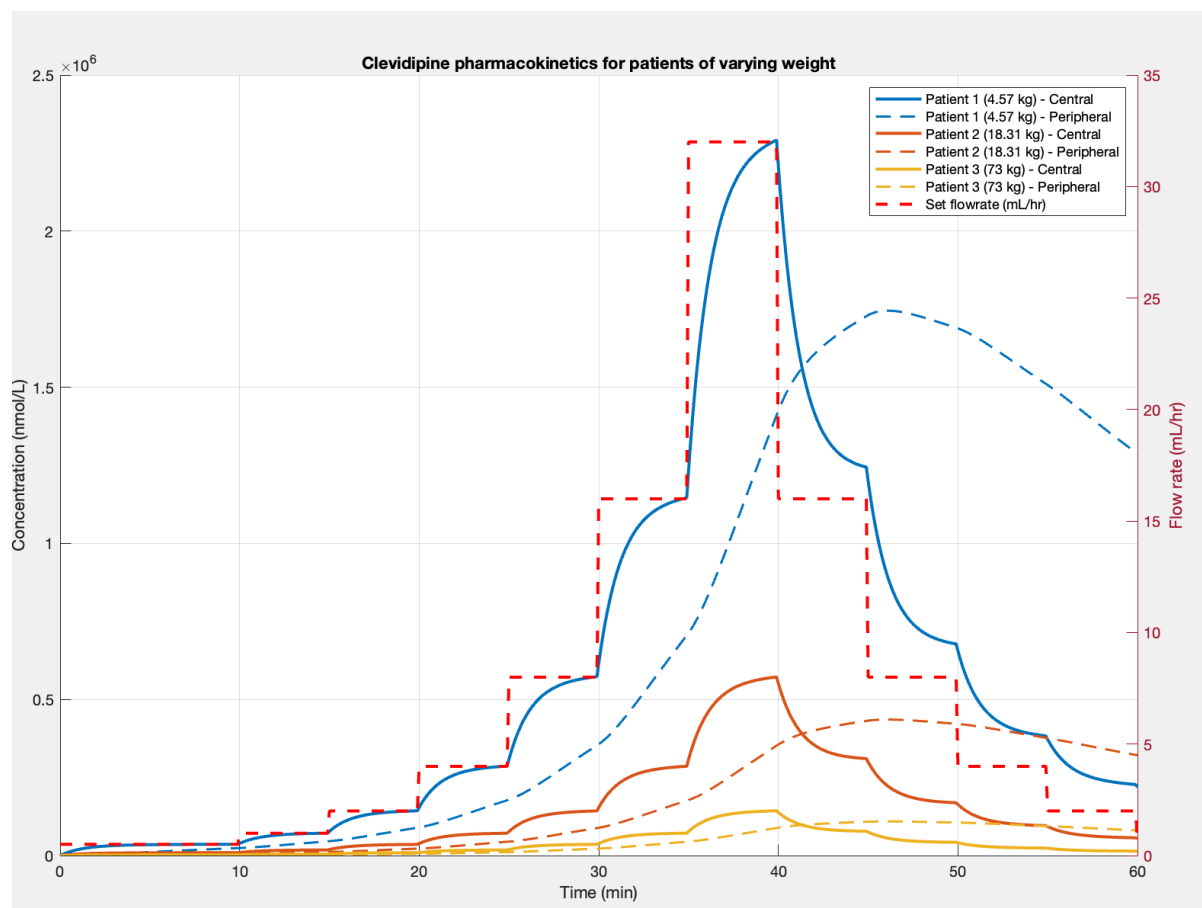


Figure 5: Clevidipine pharmacokinetics for patients of varying weight.

These effects are illustrated in the figures above. Within drug, patients with significantly smaller volumes of distribution consistently feature greater accumulation of the drug at each time step t . Between drugs, the medication with the larger half-life, nitroglycerin, consistently achieves higher concentrations of the drug over time due to its higher volume of distribution whilst also featuring a slower level of response to dosing changes due to less efficient clearance levels in Figures 4 – 6. Taken in combination, these effects demonstrate that nitroglycerin takes longer to reach steady-state concentration than clevidipine in Figure 7. This makes the patients being administered nitroglycerin slightly more vulnerable to long-term exposure and toxicity than clevidipine, particularly in patients with smaller volumes of distribution, such as infants and neonates. This effect would be even more significant for drugs with an even larger disparities in ratios of volume of distribution and blood clearance.

Clarification

For three drugs, norpinephrine, epinephrine, and vasopressin, the values of volume of distribution and blood clearance values from the studies referenced were used to choose particular half-lives that would provide us with a desired range. In the following figure, we aim to demonstrate how the selection of one half-life using the reported values instead of another, impacts the drug's accumulation rate as well as the responsiveness of its concentration to dosing. The size of this effect is influenced by the relationship between clearance and volume of distribution as discussed in the previous sections.

To begin, we will manipulate equation (5) to make clearance level more apparent, resulting in equation (19) below.

$$k = \frac{\ln(2)}{\text{half-life}} \quad (19)$$

$$\text{Volume of distribution} = \frac{\text{Clearance}}{k} \quad (20)$$

Table 1: **Drug parameters**

Name	Half-life (min)	Volume of Distribution (L/kg)	Blood Clearance Level
Norepinephrine	3	8.8	2.03 ²
Norepinephrine	4	8.8	1.52
Norepinephrine	5	8.8	1.22 ³
Epinephrine	3	0.6	0.16 ⁴
Epinephrine	8	0.6	0.052 ⁵
Epinephrine	10	0.6	0.048
Vasopressin	10	0.2	0.014 ⁶
Vasopressin	14	0.2	0.0069
Vasopressin	35	0.2	0.004 ⁷

extension

This work on plasma concentration could potentially be extended to an extremely basic argument for the efficiency of Target-Controlled Infusion (TCI), a method of titrating anesthetics which aims to optimize time to achieve steady-state concentration by algorithmically choosing the most efficient dose rate. However, the key difference is that while TCI is typically applied in context of the administration of anesthetics where effect site concentrations are both observable and predictable, our work is primarily concerned with modeling concentrations of hemodynamic agents over time; drugs which present more variability due to a greater sensitivity to the influence of patient-specific factors such as vasoreactivity and cardiac status. Furthermore, dosing for hemodynamic agents is driven largely by real-time clinical effects rather than predictable effect site concentrations readily referenceable for anesthetic drugs. Therefore our extension to TCI is limited to the demonstration of the concept that because both equations (5) and (16), which model concentration of the drug within one and two compartments respectively, are directly dependent on dose rate, guiding dose adjustments will increase concentration at each time step t and subsequently will decrease time to achieve target concentration. However these adjustments would need to be made in consideration with the principle that drugs with longer half-lives with a larger ratio of volume of distribution to clearance will sustain these concentrations for a longer period than a faster acting drug, after desired effects have been achieved.

By adjusting dose rate within the two-compartment model within the range of acceptable dose rates for the drug, one could simulate how time to reach target concentration changes with respect to dosing regimen.

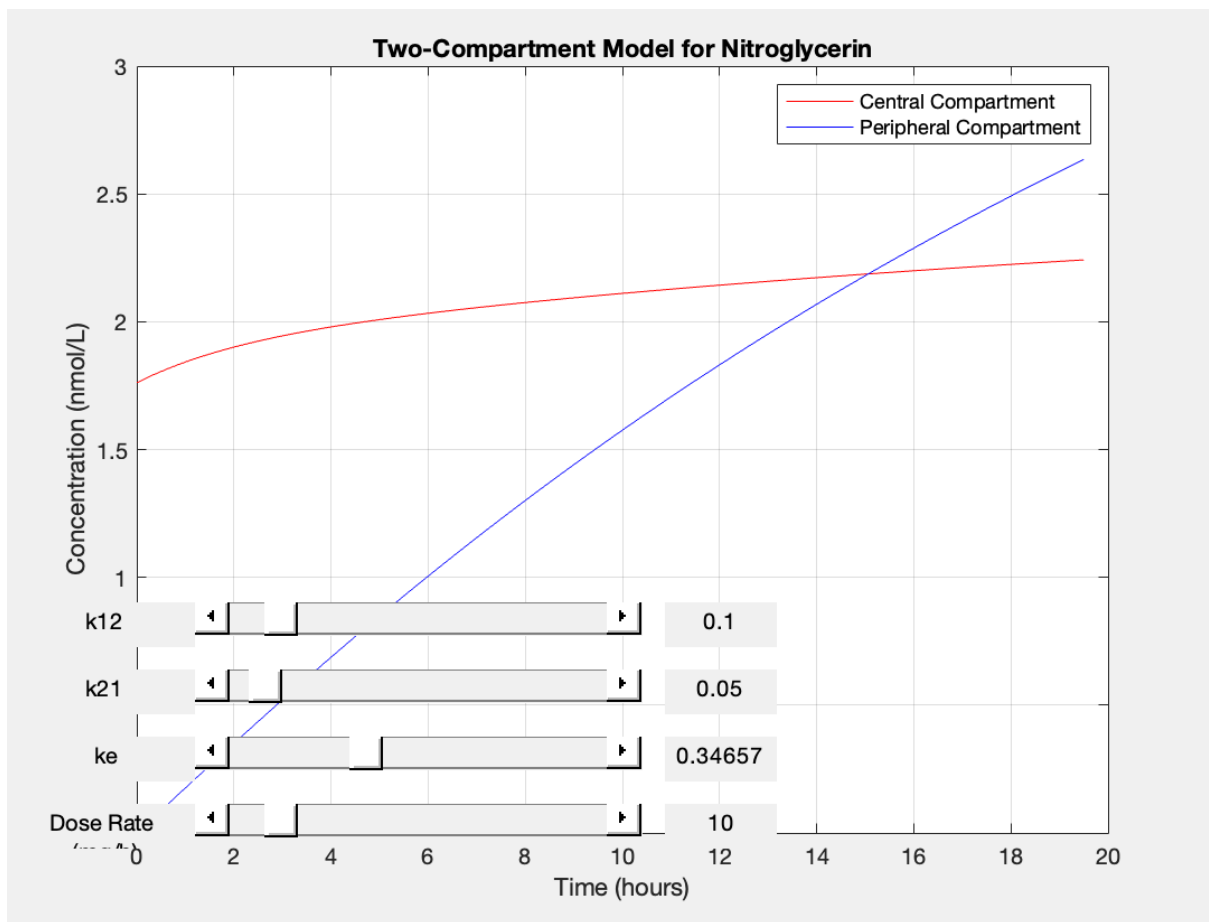


Figure 6: Adjusting titration of nitroglycerin in adult patient.

Code

<https://github.com/amalhani/multidrugcomp-pkpd>

References

- [1] Michael Argenziano et al. “Management of Vasodilatory Shock after Cardiac Surgery: Identification of Predisposing Factors and Use of a Novel Pressor Agent”. In: *The Journal of Thoracic and Cardiovascular Surgery* 116.6 (Dec. 1998). Accessed 10 Feb. 2022, pp. 973–980. DOI: 10.1016/s0022-5223(98)70049-2. URL: [https://doi.org/10.1016/s0022-5223\(98\)70049-2](https://doi.org/10.1016/s0022-5223(98)70049-2).
- [2] *Arginine Vasopressin* 1-8. <https://pubchem.ncbi.nlm.nih.gov/compound/Arginine-vasopressin-1-8>. 2019.
- [3] Marcia Buck. *PEDIATRIC PHARMACOTHERAPY Self-Injectable Epinephrine for Initial Management of Anaphylaxis in Children*. 2008.
- [4] *Clevidipine*. <https://pubchem.ncbi.nlm.nih.gov/compound/Clevidipine>. 2025.
- [5] *CLEVIPREX Dosing and Administration Video*. https://cleviprex.com/dosing-administration/?gclid=EAIaIQobChMIw7WKz9q1_QIVFR19Ch1vkQmVEAAYASAAEgKY3vD_BwE. Accessed 9 Apr. 2025. Feb. 2023.
- [6] W. E. Clutter et al. “Epinephrine Plasma Metabolic Clearance Rates and Physiologic Thresholds for Metabolic and Hemodynamic Actions in Man”. In: *Journal of Clinical Investigation* 66.1 (July 1980), pp. 94–101. DOI: 10.1172/jci109840.
- [7] Guillaume Debaty et al. “Shock-Associated Cardiac Arrest: Vasodilator Therapy May Help”. In: *American Journal of Respiratory and Critical Care Medicine* 197.7 (Apr. 2018). Accessed 9 Apr. 2025, pp. 850–852. DOI: 10.1164/rccm.201712-2596ed.
- [8] *Earlier Epinephrine Treatment during Cardiac Arrest Linked to Better Recovery*. <https://newsroom.heart.org/news/earlier-epinephrine-treatment-during-cardiac-arrest-linked-to-better-recovery>. American Heart Association Resuscitation Science Symposium 2021, Presentation 8875. 2021.
- [9] *Epinephrine*. <https://pubchem.ncbi.nlm.nih.gov/compound/Epinephrine>. 2019.
- [10] H. Ericsson et al. “Pharmacokinetics and Arteriovenous Differences in Clevidipine Concentration Following a Short- and a Long-Term Intravenous Infusion in Healthy Volunteers”. In: *Anesthesiology* 92.4 (2000), pp. 993–1001. DOI: 10.1097/00000542-200004000-00016. URL: <https://pubmed.ncbi.nlm.nih.gov/10754618/>.
- [11] H. Ericsson et al. “Pharmacokinetics and Pharmacodynamics of Clevidipine in Healthy Volunteers after Intravenous Infusion”. In: *European Journal of Clinical Pharmacology* 55.1 (1999), pp. 61–67. DOI: 10.1007/s002280050594. URL: <https://pubmed.ncbi.nlm.nih.gov/10206087/>.
- [12] Pierre Foulon and Daniel De Backer. “The Hemodynamic Effects of Norepinephrine: Far More than an Increase in Blood Pressure!” In: *Annals of Translational Medicine* 6.S1 (Nov. 2018), S25–S25. DOI: 10.21037/atm.2018.09.27. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6291609/>.
- [13] J Hilsted et al. “Whole Body Clearance of Norepinephrine. The Significance of Arterial Sampling and of Surgical Stress”. In: *The Journal of Clinical Investigation* 71.3 (1983), pp. 500–505. DOI: 10.1172/jci110794. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC436897/>.

- [14] Kyle H. Kim and Derek J. Schaller. *Nitroglycerin*. <https://www.ncbi.nlm.nih.gov/books/NBK482382/>. 2023.
- [15] E. F. McNiff et al. “Nitroglycerin Pharmacokinetics after Intravenous Infusion in Normal Subjects”. In: *Journal of Pharmaceutical Sciences* 70.9 (1981). Accessed 28 Nov. 2022, pp. 1054–1058. DOI: 10.1002/jps.2600700923.
- [16] Jayanta Mitra et al. “Vasopressin: Its Current Role in Anesthetic Practice”. In: *Indian Journal of Critical Care Medicine* 15.2 (2011), pp. 71–77. DOI: 10.4103/0972-5229.83006.
- [17] *Nitroglycerin*. <https://pubchem.ncbi.nlm.nih.gov/compound/Nitroglycerin>. 2019.
- [18] *Noradrenaline (Critical Care Formulary) | Right Decisions*. <https://rightdecisions.scot.nhs.uk/>
- [19] *Norepinephrine*. <https://pubchem.ncbi.nlm.nih.gov/compound/Norepinephrine>. 2019.
- [20] *Norepinephrine (Intravenous Route) Description and Brand Names - Mayo Clinic*. <https://www.mayoclinic.org/drugs-supplements/norepinephrine-intravenous-route/description/drg-20490844>.
- [21] Mehdi Oualha et al. “Population Pharmacokinetics and Haemodynamic Effects of Norepinephrine in Hypotensive Critically Ill Children”. In: *British Journal of Clinical Pharmacology* 78.4 (2014), pp. 886–897. DOI: 10.1111/bcp.12412. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4239982/>.
- [22] F. Estelle R. Simons et al. “Epinephrine Absorption in Children with a History of Anaphylaxis”. In: *Journal of Allergy and Clinical Immunology* 101.1 (1998), pp. 33–37. DOI: 10.1016/s0091-6749(98)70190-3.
- [23] *VASOPRESSIN INJECTION USP 20 UNITS per ML 1mL VIAL*. <https://dailymed.nlm.nih.gov/dailymed/fda/fdaDrugXsl.cfm?setid=f5f03364-e71c-9bcd-e053-2995a90a8ada&type=display>.
- [24] James Waterson et al. “Pandemic Response- Maintaining Effective and Safe IV Infusion Therapy for Critically Ill and Isolated Patients”. In: *Hospitals* (2020), pp. 28–31. URL: https://www.researchgate.net/publication/342887301_Pandemic_Response-_Maintaining_Effective_and_Safe_IV_Infusion_Therapy_for_Critically_Ill_and_Isolated_Patients.